

Updated Recommendations

Following testing, the Week 6 recommendations were retained but refined to ensure that they accurately reflected the evidence.

1. Evaluate and strengthen the reminder process

Reminder coverage was 72.68%, while the no-reminder group had a higher observed no-show rate. HealthConnect should investigate reminder delivery and coverage, while avoiding a causal interpretation of the difference.

2. Prioritise patients with previous no-show behaviour

Previous attendance behaviour can be considered when developing future risk segmentation or targeted attendance-support strategies.

3. Investigate accessibility for patients travelling longer distances

The 20+ km group recorded a 57.29% no-show rate. Further investigation could determine whether travel distance or accessibility presents an operational barrier.

4. Improve waiting-time data quality

The waiting-time field contains missing information, making the Unknown category difficult to interpret. Improving data completeness should be prioritised before using this variable for stronger operational conclusions.

5. Develop a future no-show risk model

The analysis identified candidate variables that could be assessed by Data Science, including:

- Previous no-shows
- Previous appointments
- Reminder status
- Distance to clinic
- Waiting time
- Appointment type
- Age group
- Booking lead days

These should be treated as **candidate modelling features**, rather than confirmed predictors.

Cross-Track Testing Contribution

The Analytics track's main cross-track contribution was to provide **validated analytical findings and candidate variables** that could support the planned Data Science no-show prediction use case.

The validated findings relevant to Data Science include:

- Overall no-show behaviour
- Previous no-show history
- Reminder status
- Distance to clinic
- Waiting time
- Appointment type
- Age group
- Booking lead time

These findings can help Data Science assess potential features for future predictive modelling.

The Analytics team also identified data-quality considerations, particularly missing waiting-time and distance values, which should be considered during modelling and preprocessing.

A fully completed two-way Analytics–Data Science validation cycle could not be documented because of limited availability and responsiveness from participants remaining active in the Data Science track. Therefore, the contribution is documented as a **prepared and validated analytical input**, rather than claiming completed model integration that was not evidenced.

Remaining Analytical Issues

The following issues remain open after Week 7 testing:

Issue	Impact	Recommended Action
Missing waiting-time data	Limits interpretation of waiting-time patterns	Improve data capture/completeness
Missing distance data	Limits analysis of accessibility	Investigate source/system causing missing values
Small high-risk segments	Percentages may be unstable	Use minimum sample thresholds
Observational relationships	Cannot establish causality	Conduct further statistical/predictive analysis
No validated predictive model	Risk groups remain descriptive	Data Science to develop and validate model
Cross-track integration	No completed two-way validation	Continue Analytics–Data Science collaboration
Dashboard interpretation	Some findings could be overinterpreted	Maintain clear labels and explanatory notes

Week 8 Recommendations

For Week 8, the Analytics track should focus on **final integration, decision support and preparation for future predictive analysis**.

Recommended priorities

1. Finalise the validated dashboard

Ensure that the refined dashboard contains the validated KPIs, clear labels, appropriate filters and explanatory notes for missing data and observational findings.

2. Maintain a clear distinction between association and causation

Future reporting should avoid statements such as "reminders reduce no-shows" unless causal evidence becomes available.

3. Improve data quality

Prioritise completeness and consistency of:

- Waiting-time data
- Distance-to-clinic data
- Reminder information

4. Support Data Science modelling

Provide the validated candidate variables and analytical findings to Data Science for potential no-show prediction modelling.

5. Validate predictive outputs when available

If Data Science produces a model, Analytics should assess whether the model outputs are understandable, consistent with the underlying data and suitable for dashboard/decision-support use.

6. Continue cross-track integration

Establish a clearer feedback loop between Analytics, Data Science and Project Management so that analytical findings can be translated into tested operational recommendations.

7. Move from descriptive analysis toward decision support

The validated Week 7 findings can form the foundation for future analysis that helps HealthConnect identify potentially higher-risk appointments and determine where operational interventions may warrant further investigation.

Final Week 7 Conclusion

Week 7 testing confirmed that the core analytical outputs from Week 6 were accurate and consistent with the underlying HealthConnect appointment dataset. The overall no-show rate of 48.46%, higher observed no-show rates among patients travelling 20+ km, differences across reminder-status groups and the relationship between previous no-show history and current attendance were all supported by validation.

Testing also identified important analytical limitations, particularly missing data, small segment sizes and the observational nature of the analysis. As a result, several recommendations were refined to avoid implying causation where the data only demonstrates association.

The dashboard was refined to improve clarity, interpretation and usability, and the key outputs were re-tested after refinement. The validated findings and candidate variables can support future Data Science modelling and wider HealthConnect decision-making, while remaining data-quality and cross-track issues should continue to be addressed in Week 8.